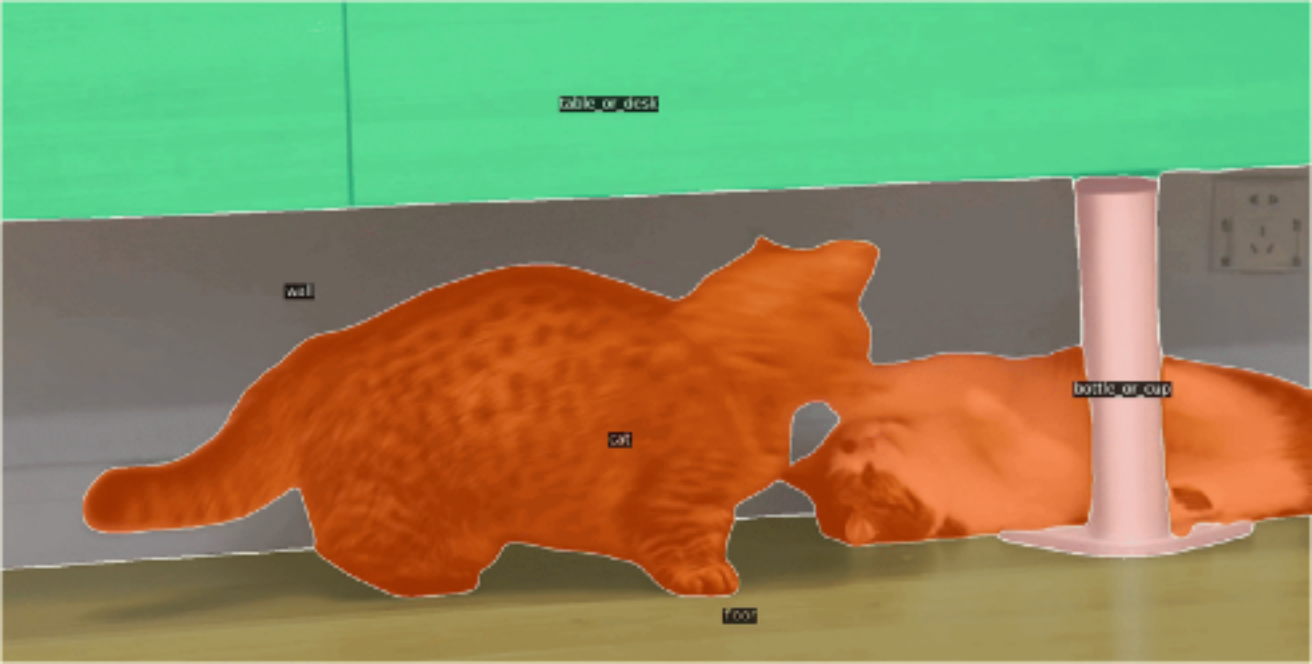

table_or_desk
wall
cat
bottle_or_cup
floor

table or desk
wall
cat
cat
bottle or cup
floor

Video Instance Segmentation (VIS)

Video Semantic Segmentation (VSS)

Video Panoptic Segmentation (VPS)

Video Object Segmentation (VOS)

Visual prompt

Text prompt

A cat quickly darting across the camera.

Prompt-guided Segmentation

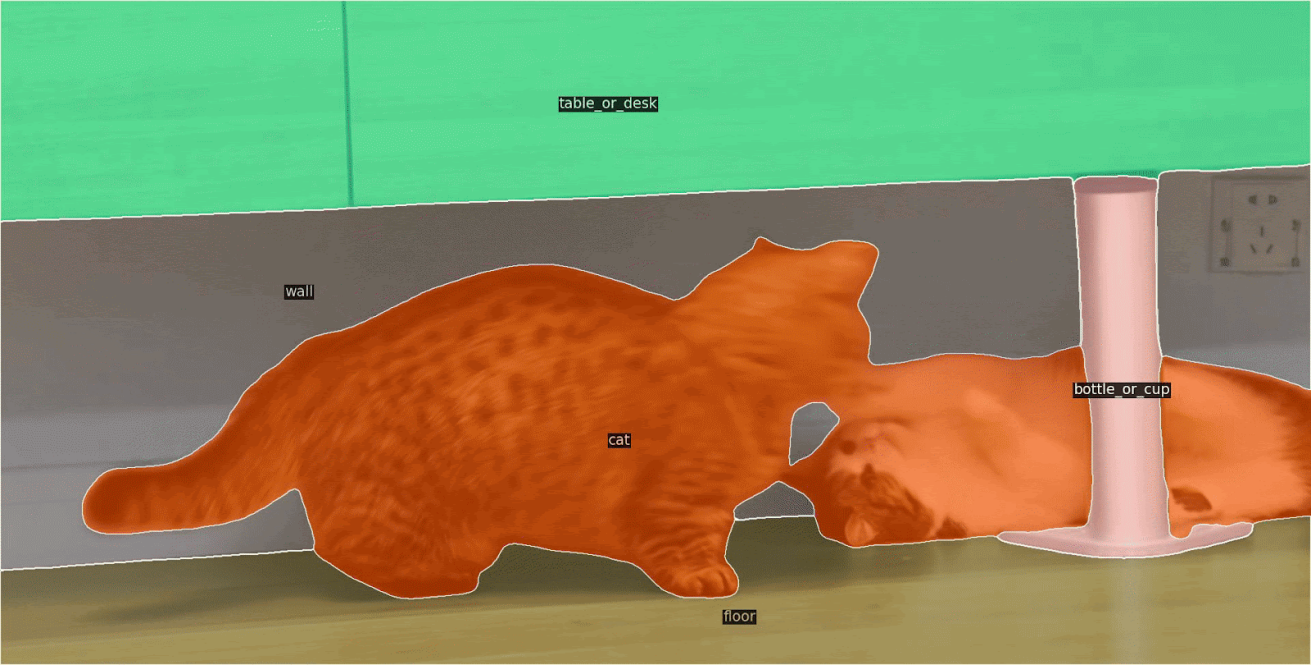
Category-guided Segmentation

Consistent understanding across different tasks

UniVS: Video Segmentation Foundation Model

Minghan Li. **UniVS:** Unified Video Segmentation with Prompts (CVPR 2024)

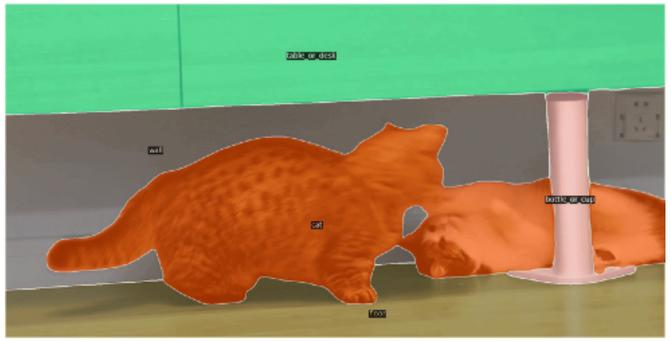
Cat 87%
Cat 87%


table_or_desk
wall
cat
bottle_or_cup
floor

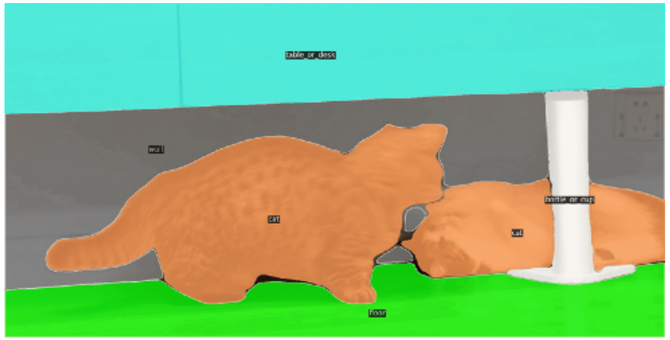
table_or_desk
wall
cat
cat
bottle_or_cup
floor

UniVS: Video Segmentation Foundation Model

Video Instance Segmentation (VIS)



Video Semantic Segmentation (VSS)



Video Panoptic Segmentation (VPS)

Visual prompt

Text prompt

A cat quickly darting across the camera.

Video Object Segmentation (VOS)

Category-guided Segmentation

Prompt-guided Segmentation

Consistent understanding across different tasks

Minghan Li. **UniVS:** Unified Video Segmentation with Prompts (CVPR 2024)

UniVS: Video Segmentation Foundation Model

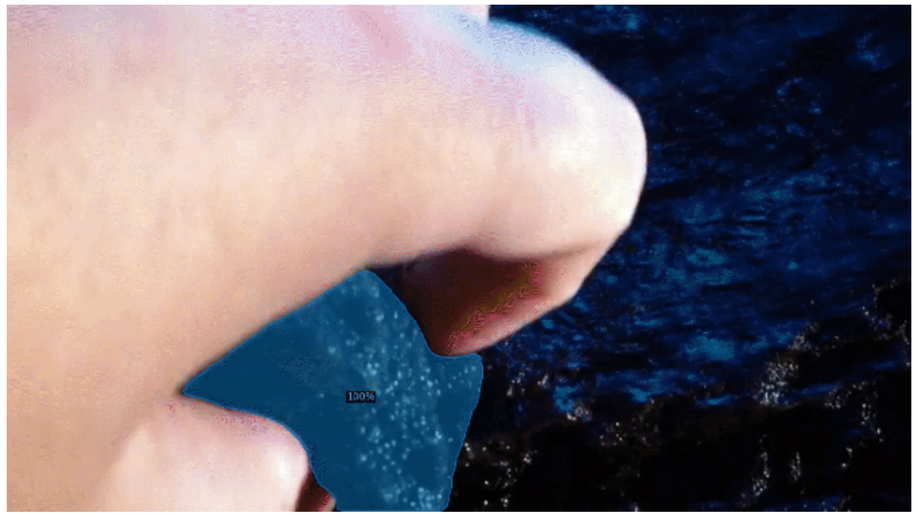
Prompt-guided Segmentation

Text description (open-world voc.)

The brown kangaroo is on the green grass looking behind

Text description (open-world voc.)

The frog is held by a person in his hand and place near the another frog



Better generalization & robustness in challenging scenarios

Minghan Li. **UniVS:** Unified Video Segmentation with Prompts (CVPR 2024)